

Ruchita Kailas Vehale

Bristol, UK • ruchita.uob.2024@gmail.com • +44 7818970209 • [LinkedIn](#) • [Portfolio](#) • [GitHub](#)

Professional Summary

R&D Engineer with 5+ years of experience in computer vision, robotics, autonomous systems, and embedded engineering. Skilled in machine learning, visual SLAM, drone systems, 3D localisation, and deep learning for real-world industrial and robotic applications. Published researcher in Wiley and RSC journals with expertise spanning AI-driven perception, autonomous navigation, and robotics software development. MSc Robotics graduate from the University of Bristol seeking opportunities in computer vision, robotics software, and applied AI/ML engineering.

Experience

DeGould Ltd

Bristol, UK

Associate Software Engineer (R&D Team)

October 2025 – Present

- Developed end-to-end localisation solutions for automotive defect detection systems, ensuring accurate identification and positioning of defects across multiple vehicle inspection pipelines
- Designed a Blender-based 3D localisation plugin replacing traditional 2D colour-based approaches, reducing localisation setup time from 2 days to under 1 hour (~96% faster) across 2–3 inspection pipelines
- Built computer vision and ML algorithms for pose estimation, calibration, defect localisation, and 3D–2D mesh projection workflows
- Worked with synthetic data for segmentation to improve model performance and automation
- Automated and streamlined inspection pipelines by improving codebase structure, deployment workflows, and reducing manual intervention
- Enhanced scalability, reliability, and maintainability of production localisation systems across different inspection environments

HVN Labs, Future Space, BRL

Bristol, UK

Jr. Drone System Engineer

June 2025 – August 2025

- Developed an autonomous precision landing system using YOLOv8-based drone detection on a Raspberry Pi camera, combined with multi-object tracking and 3D position estimation using Kalman filtering, Hungarian assignment, and stereo vision-based depth reconstruction for GPS-denied closed-loop landing control
- Built a supporting ESP32-based wireless telemetry module enabling real-time communication between ground systems and the drone for command and status exchange
- Integrated smart servos with an ArduPilot-based flight controller using Lua scripting, implementing control, monitoring, and fault recovery logic, and extending firmware for native CAN/UART support

Technology Intern

April 2025 – May 2025

- Contributed to development and integration of a coaxial drone platform for synchronized lighting displays, including ESC calibration, flight controller setup, and flight testing
- Prototyped a vision-assisted GPS fusion landing system to improve landing accuracy and repeatability in dynamic outdoor environments

CMET

Pune, India

Research Fellow

April 2023 - Aug 2024

- Optimized multi-band antenna designs using CST Microwave Studio, achieving ~20% size reduction and ~70% improvement in simulation accuracy across GPS, Wi-Fi, Bluetooth, and 5G frequency bands
- Performed antenna prototyping and RF characterization using FDM/inkjet 3D printing, Vector Network Analyzer (VNA), and X-ray Diffraction (XRD) for design validation and performance analysis

Project: Object detection & velocity estimation using for Defence application

- Built a MATLAB computer vision system for multi-object detection, tracking, and velocity estimation using GMM background subtraction, Kalman filtering, and Hungarian algorithm — with CLAHE and Hough Transform-based image enhancement for high-intensity lighting conditions
- Implemented camera calibration and projective geometry-based scaling to convert pixel motion into real-world velocity measurements, validated across multiple video datasets

Projects

1. Dynamic Object Aware Visual SLAM

- Developed dynamic object-aware RGB-D and monocular SLAM systems using Intel RealSense D455, Jetson Orin Nano, and iPhone RGB cameras, integrating YOLOv8, DeepSORT, and MiDaS for real-time object detection, tracking, and depth estimation
- Implemented RGB-D and estimated-depth dynamic filtering with inpainting-based background reconstruction, achieving 94.8% human pixel removal, and applied ORB feature matching with pose graph optimization for loop closure and drift correction
- Built a complete real-time mapping pipeline with RANSAC-based pose estimation and SE(3) optimization, generating 2D occupancy grids, 3D point clouds (PLY), and stable trajectory outputs at 3.5+ FPS in dynamic indoor environments.

2. DriveSense: Real-Time OOD Detection & VLM Hazard Reasoning for Autonomous Driving (*In Progress*)

- Developing a real-time out-of-distribution (OOD) detection pipeline for autonomous driving safety using SegFormer-B2, DINOv2, and anomaly-aware segmentation on Cityscapes and Fishyscapes benchmarks
- Implementing multiple OOD scoring methods including MSP, entropy-based uncertainty, feature-bank KNN (FAISS), and synthetic anomaly augmentation with evaluation via AUROC, AUPRC, and FPR@95TPR
- Building a Vision-Language hazard reasoning layer using Claude/Qwen-VL models to generate structured hazard reports for downstream planner integration
- Deploying the pipeline with ONNX Runtime, FastAPI, Docker, Streamlit, CI/CD, and cloud training workflows on GCP Vertex AI

3. Mobile Robot Navigation and Leader-Follower Control

- Developed autonomous mobile robot navigation using PID control, boundary detection, and shortest-path planning for target localization
- Created a robotic leader follower system with dynamic speed control and sensor fusion, reducing tracking errors by 97.93% for autonomous and swarm robotics applications using QTR-RC sensors

4. AI Dungeon Maze Environment

- Built a custom Gymnasium-based maze environment in Python implementing supervised (PCA-KNN), unsupervised (GMM clustering), and reinforcement learning (Q-Learning, Monte Carlo, Temporal Difference, Policy Iteration) techniques for autonomous agent navigation, with Pygame-based visualisation and convergence diagnostics

Publications

1. [Binder Jet 3D Printed Ceramic Substrates for 5G Patch Antenna](#)

Ruchita K. Vehale, *et al.*, *Advanced Materials Technologies* (Wiley)

2. [Comparative Analysis of Conventional and Deep Learning Algorithms for Apple Detection](#)

Ruchita K. Vehale, *et al.*, *SSRN Electronic Journal* (Elsevier), Sep 2025

3. [Preparation of Phase-Pure MgTiO₃ Microwave Dielectric Ceramics for GPS Antenna Application](#)

Ruchita K. Vehale, *et al.*, *Journal of Materials Chemistry C* (RSC), January 2024

4. [Fabrication of 3D Printable Low-Loss, Dielectric-Enhanced ABS–ZnO Nanocomposite Filaments for Microstrip Patch Antenna in GPS Communication](#)

Ruchita K. Vehale, *et al.*, *Journal of Polymer Engg and Science* (Wiley), April 2026

Education

University of Bristol

Master of Science, Robotics

Bristol, UK

Aug 2024 - Sep 2025

Savitribai Phule Pune University (SPPU)

Master of Technology, Electronics & Electrical | **GPA:** 9.43/10 (Gold Medallist)

Pune, India

Jun 2020 - Sep 2022

AISSMS College of Engineering, SPPU

Bachelor of Engineering, Electronics Engineering | **GPA:** 8.31/10

Pune, India

May 2016 - Nov 2020

Skills and Certifications

- **Programming:** Python, C/C++, Embedded C, MATLAB
- **Computer Vision & AI:** OpenCV, PyTorch, YOLOv8, DeepSORT, MiDaS, Scikit-learn, Keras
- **Robotics & Embedded Systems:** Stereo Vision, Jetson, Visual SLAM, LiDAR, Raspberry Pi, ESP32, ArduPilot
- **Design & Simulation:** Blender, SolidWorks, CST Microwave Studio, Choregraphe
- **Cloud & Tools:** AWS, Linux, Github
- **Certifications:** Python for Deep Learning Applications; Mechatronics in PLC and SCADA

Leadership & Activities

National Cadet Corps (Air Force)

May 2017 - Feb 2020

- NCC Cadet Sergeant with leadership of 500-member teams; awarded Best Cadet and Gold Medal in shooting

Language: English, Hindi, Marathi, Japanese (N4 & N5 level)

Interests: Tabla (Indian classical percussion), rifle shooting, painting, tree plantation